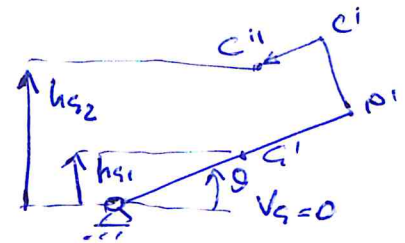


$$[\Lambda_m(\underline{x}_0)] = \begin{bmatrix} L & 0 \\ 1 & 0 \\ -AP \sin \vartheta_0 - R \cos \vartheta_0 & -R \cos \vartheta_0 \\ AP \cos \vartheta_0 - R \sin \vartheta_0 & -R \sin \vartheta_0 \\ 1 & 1 \end{bmatrix} \quad \underline{x}_0 = \begin{bmatrix} \vartheta_0 \\ 0 \end{bmatrix}$$

$$[\Lambda_v(\underline{x}_0)] = [\Lambda_c(\underline{x}_0)] = \begin{bmatrix} 2L \cos \vartheta_0 & 0 \\ AP \sin \vartheta_0 + R \cos \vartheta_0 & R \cos \vartheta_0 \end{bmatrix}$$

$$[\Lambda_d(\underline{x}_0)] = \begin{bmatrix} AP \cos \vartheta_0 - R \sin \vartheta_0 & -R \sin \vartheta_0 \\ 2L \cos \vartheta_0 & 0 \end{bmatrix}$$

$$\begin{cases} h_{s1} = L \sin \vartheta \\ h_{s2} = y_{c''} = AP \sin \vartheta + R \cos \vartheta - R y_r \sin \vartheta \end{cases}$$



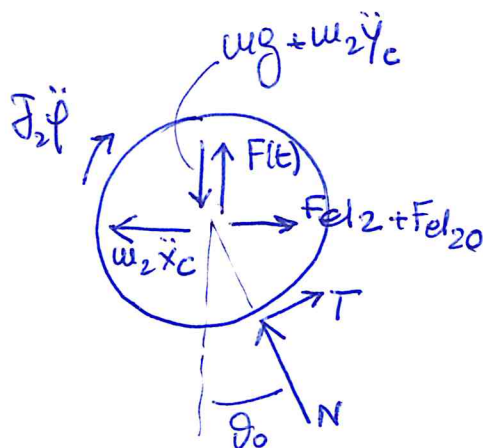
$$\begin{cases} \frac{\partial^2 h_{s1}}{\partial \vartheta^2} = -L \sin \vartheta \\ \frac{\partial^2 h_{s2}}{\partial \vartheta^2} = -AP \sin \vartheta - R \cos \vartheta + R y_r \sin \vartheta \\ \frac{\partial^2 h_{s2}}{\partial \vartheta \partial \varphi} = -R \cos \vartheta \\ \frac{\partial^2 h_{s2}}{\partial \varphi^2} = 0 \end{cases}$$

$$[k_{s1}] = M_1 g \begin{bmatrix} -L \sin \vartheta_0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$[k_{s2}] = M_2 g \begin{bmatrix} -AP \sin \vartheta_0 - R \cos \vartheta_0 & -R \cos \vartheta_0 \\ -R \cos \vartheta_0 & 0 \end{bmatrix}$$

$$\begin{cases} \frac{\partial^2 \Delta L_1}{\partial \theta^2} = -2L \sin \theta \\ \frac{\partial^2 \Delta L_2}{\partial \theta^2} = AP \cos \theta + R \sin \theta - R \sin \theta \\ \frac{\partial^2 \Delta L_2}{\partial \theta \partial \varphi} = -R \sin \theta \\ \frac{\partial^2 \Delta L_2}{\partial \varphi^2} = 0 \end{cases} \quad \begin{aligned} (K_{II,1}) &= k_1 \Delta L_{01} \begin{bmatrix} -2L \sin \theta_0 & 0 \\ 0 & 0 \end{bmatrix} \\ (K_{II,2}) &= k_2 \Delta L_{02} \begin{bmatrix} AP \cos \theta_0 - R \sin \theta_0 & -R \sin \theta_0 \\ -R \sin \theta_0 & 0 \end{bmatrix} \end{aligned}$$

question nr. 2



$$F_{el2} = (k_2 + i \omega c_2) \Delta L_{2,lin}$$

$$F_{el20} = k_2 \Delta L_{02}$$

$$\Delta L_{2,lin} = (AP \sin \theta_0 + R \cos \theta_0) (\theta - \theta_0) + R \cos \theta_0 \varphi_r$$

$$\ddot{x}_{c,lin} = -\omega^2 (-AP \sin \theta_0 - R \cos \theta_0) (\theta - \theta_0) - R \cos \theta_0 \varphi_r$$

$$\ddot{y}_{c,lin} = -\omega^2 (L_0 \cos \theta_0 - R \sin \theta_0) (\theta - \theta_0) - R \sin \theta_0 \varphi_r$$

$$\sum \vec{F}_N = 0: N - F_{el20} \sin \theta_0 - F_{el2} \sin \theta_0 + \omega_2 \ddot{x}_c \sin \theta_0 + (F(t) - \omega_2 g - \omega_2 \ddot{y}_c) \cos \theta_0 = 0 \rightarrow$$

$$\rightarrow N = \underbrace{(F_{el20} \sin \theta_0 + \omega_2 g \cos \theta_0)}_{N_{ST}} + \underbrace{[F_{el2} \sin \theta_0 - \omega_2 \ddot{x}_c \sin \theta_0 + (F(t) - \omega_2 \ddot{y}_c) \cos \theta_0]}_{N_{dyn}} = N_{ST} + N_{dyn}$$

loss of contact when $N = 0 \rightarrow N_{ST} + N_{dyn} = 0$